

JABstem

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Past Papers

Nat 5

Physics

2023 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	/100	%	
A	65+	65%	34.7%
B	52+	52%	19.8%
C	40+	40%	16.4%
D	27+	27%	15.5%
No award	<27	<27%	13.6%

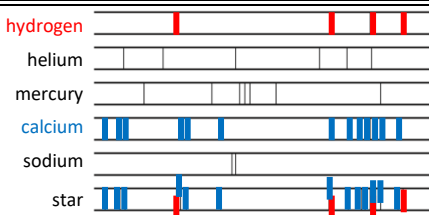
Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	14.5 /25	38.8 /75	No Assignment in 2023

2023 Nat5 Physics Marking Scheme

Question	Answer	% Correct	Physics Covered																											
1	D	82	<table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>Vectors</td><td>force</td><td>acceleration</td></tr><tr><td>Vectors have both magnitude and direction while scalars only have magnitude</td><td>Force is a vector quantity with magnitude and direction.</td><td>Acceleration is a vector quantity with magnitude and direction.</td></tr></table>	X	Y	Z	Vectors	force	acceleration	Vectors have both magnitude and direction while scalars only have magnitude	Force is a vector quantity with magnitude and direction.	Acceleration is a vector quantity with magnitude and direction.																		
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2	D	44	☒ A $t_1 + t_2$ does not measure the time taken for the card to pass through light gate Q ☒ B $t_1 + t_2$ does not measure the time taken for the card to pass through light gate Q ☒ C this would measure the instantaneous speed through light gate P not light gate Q ☒ D this measures the instantaneous speed through light gate Q ☒ E this does not measure the instantaneous speed at any point in the experiment																											
			3	C	62	<table><tr><th>Area ❶</th><th>Area ❷</th><th>Area ❸</th></tr><tr><td>Distance = area under graph</td><td>Distance = area under graph</td><td>Distance = area under graph</td></tr><tr><td>$= 2.0 \times 4.0$</td><td>$= \frac{1}{2} \times 6.0 \times 2.0$</td><td>$= 6.0 \times 4.0$</td></tr><tr><td>$= 8\text{m}$</td><td>$= 6\text{m}$</td><td>$= 24\text{m}$</td></tr><tr><td colspan="3">Total Distance = $8 + 6 + 24 = 38\text{m}$</td></tr></table>	Area ❶	Area ❷	Area ❸	Distance = area under graph	Distance = area under graph	Distance = area under graph	$= 2.0 \times 4.0$	$= \frac{1}{2} \times 6.0 \times 2.0$	$= 6.0 \times 4.0$	$= 8\text{m}$	$= 6\text{m}$	$= 24\text{m}$	Total Distance = $8 + 6 + 24 = 38\text{m}$											
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4	D	28	$E_w = ?$ $F = 6.0\text{N}$$d = 3.0\text{m}$ $E_w = F \times d$$E_w = 6.0 \times 3.0$$E_w = 18\text{ J}$																											
			5	B	40	Kinetic energy at 2.0 m s^{-1} $E_k = ?$ $m = 4.0\text{ kg}$$v = 2.0\text{ m s}^{-1}$ $E_k = \frac{1}{2} \times m \times v^2$$E_k = \frac{1}{2} \times 4.0 \times (2.0)^2$$E_k = \frac{1}{2} \times 4.0 \times 4.0$$E_k = 8.0\text{ J}$ Kinetic energy at 6.0 m s^{-1} $E_k = ?$ $m = 4.0\text{ kg}$$v = 6.0\text{ m s}^{-1}$ $E_k = \frac{1}{2} \times m \times v^2$$E_k = \frac{1}{2} \times 4.0 \times (6.0)^2$$E_k = \frac{1}{2} \times 4.0 \times 36$$E_k = 72.0\text{ J}$ Increase in kinetic energy = $72.0\text{ J} - 8.0\text{ J} = 64\text{ J}$																								
						6	D	58	$T = ?$ $\alpha = 0.290$$d = 1\text{ AU}$ <table><tr><td>$T^2 =$</td><td>280^2</td><td>$\times$</td><td>$\sqrt{\frac{(1-\alpha)}{d^2}}$</td></tr><tr><td>$T^2 =$</td><td>$78400$</td><td>$\times$</td><td>$\sqrt{\frac{(1-0.290)}{1^2}}$</td></tr><tr><td>$T^2 =$</td><td>$78400$</td><td>$\times$</td><td>$\sqrt{\frac{0.71}{1}}$</td></tr><tr><td>$T^2 =$</td><td>$78400$</td><td>$\times$</td><td>$0.843$</td></tr><tr><td>$T^2 =$</td><td>$66061$</td><td></td><td></td></tr><tr><td>$T =$</td><td>257 K</td><td></td><td></td></tr></table>	$T^2 =$	280^2	$\times$	$\sqrt{\frac{(1-\alpha)}{d^2}}$	$T^2 =$	78400	$\times$	$\sqrt{\frac{(1-0.290)}{1^2}}$	$T^2 =$	78400	$\times$	$\sqrt{\frac{0.71}{1}}$	$T^2 =$	78400	$\times$	0.843	$T^2 =$	66061			$T =$
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7	A	67	☒ A Asteroid – object orbiting a star which is smaller than a dwarf planet ☒ B Dwarf Planet – object which orbits a star but not large enough to be classed as a small planet ☒ C Exoplanet – planet orbiting around a star outside our solar system ☒ D Planet – large ball of matter. Orbiting a star which does not emit light ☒ E Star – large ball of matter undergoing nuclear fusion emitting light and EM radiation						
8	D	48	Unwards force = 2200N Downwards force = mg = 350kg x 3.7 N kg ⁻¹ = 1295 N Unbalanced force = 2200N – 1295N = 905N upwards As space vehicle is decending when rockets fire then the vehicle will still decend to the surface but there is a decrease in speed as there is a unbalanced force upwards.						
9	E	79	☒ A A negative particle P will deflect away from a negatively charged plate R ☒ B If particle P had no charge it would pass through undeflected through electric field ☒ C If particle P had no charge it would pass through undeflected through electric field ☒ D A positive particle P will deflect away from a positively charged plate R ☒ E A positive particle P will bend towards a negatively charged plate R						
10	C	81	☒ A Negative charges (electrons) move in both a.c. and d.c. current ☒ B In a.c. current, direction of current reverses constantly back and forth ☒ C In a.c. current, charges (electrons) reverse intervals at regular interval ☒ D Only negative charges (electons) move in a current in both a.c. and d.c. current ☒ E The quantity of current rises and falls during a.c. current						
11	D	35	Voltage across resistor R = 5.0V – 2.2V = 2.8V Current in Resistor R = 10.0 mA = 0.0100 A $R = \frac{V}{I} = \frac{2.8}{0.0100} = 280 \Omega$						
12	A	58	$V_s = 24 \text{ V}$ $V_2 = ?$ $R_1 = 2.4 \text{ k}\Omega$ $R_2 = 1.2 \text{ k}\Omega$ $V_2 = \frac{R_2}{R_1 + R_2} \times V_s$ $V_2 = \frac{1.2}{2.4 + 1.2} \times 24$ $V_2 = 8 \text{ V}$						
13	C	44	☒ A Variable resistor is at bottom of circuit so motor will switch on when light is high. ☒ B circuit contains thermistor which would make the circuit dependent on temperature ☒ C Variable resistor is at top of circuit and circuit contains LDR to switch on motor on low light ☒ D circuit contains thermistor which would make the circuit dependent on temperature not light ☒ E circuit contains thermistor which would make the circuit dependent on temperature not light						
14	E	70	$P = 250 \text{ W}$ $E = ?$ $t = 2 \text{ hours} = 2 \times 60 \times 60 \text{ s}$ $P = \frac{E}{t} \therefore 250 = \frac{E}{2 \times 60 \times 60} \therefore E = 250 \times 2 \times 60 \times 60 = 1860000 \text{ J}$						
15	C	64	From graph: When Power P = 1 W then Resistance R = 4 Ω $P = \frac{V^2}{R} \therefore 1 = \frac{V^2}{4} \therefore V^2 = 4$ $\therefore V = 2 \text{ V}$						
16	C	64	<table><tr><td>Statement I - Correct</td><td>Statement II - Correct</td><td>Statement III - Incorrect</td></tr><tr><td>The mass of the copper block gives the mass in the equation $E_h = cm\Delta T$</td><td>The initial and final temperatures are needed to to calculate ΔT for the equation $E_h = cm\Delta T$</td><td>The power supply does not give the energy supplied. Joulemeter provides E_h for the equation $E_h = cm\Delta T$</td></tr></table>	Statement I - Correct	Statement II - Correct	Statement III - Incorrect	The mass of the copper block gives the mass in the equation $E_h = cm\Delta T$	The initial and final temperatures are needed to to calculate ΔT for the equation $E_h = cm\Delta T$	The power supply does not give the energy supplied. Joulemeter provides E_h for the equation $E_h = cm\Delta T$
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17	B	57	$E = ?$ $m = 3.5 \text{ kg}$ $l = 3.34 \times 10^5 \text{ J kg}^{-1}$ $E = m \times l$ $E = 3.5 \times 3.34 \times 10^5$ $E = 1.2 \times 10^6 \text{ J}$						
18	B	65	$P = 2.0 \times 10^8 \text{ Pa}$ $F = 5.0 \text{ kN} = 5000 \text{ N}$ $A = ?$ $P = \frac{F}{A}$ $2 \times 10^8 = \frac{5000}{A}$ $A = 2.5 \times 10^{-5} \text{ m}^2$						
19	A	56	☒ A Pressure & temperature are directly proportional in <i>pressure-temperature Law calculations</i> ☒ B Pressure & temperature are directly proportional in <i>pressure-temperature Law calculations</i> ☒ C The volume is constant in the bhooy ☒ D Temperatures used in <i>pressure-temperature Law calculations</i> are measured in Kelvin ☒ E Temperatures used in <i>pressure-temperature Law calculations</i> are measured in Kelvin						
20	C	48	$p_1 = 5.0 \times 10^5 \text{ Pa}$ $p_2 = 5.5 \times 10^5 \text{ Pa}$ $V_1 = 2.2 \text{ m}^3$ $V_2 = ?$ $T_1 = 27^\circ\text{C} = 320 \text{ K}$ $T_2 = 54^\circ\text{C} = 370 \text{ K}$ $\frac{p_1 V_1}{T_1} = \text{constant} = \frac{5.0 \times 10^5 \times 2.2}{320} = 3437.5$ $\frac{p_2 V_2}{T_2} = \text{constant} = \frac{5.5 \times 10^5 \times V_2}{370} = 3437.5$ $V_2 = \frac{3437.5 \times 370}{5.5 \times 10^5}$ $V_2 = 2.3 \text{ m}^3$						
21	B	58	<table><tr><th>Amplitude</th><th>Frequency</th></tr><tr><td>Amplitude = $\frac{0.4}{2} = 0.2 \text{ m}$</td><td>Wavelength = $\frac{12 \text{ m}}{2} = 6 \text{ m}$ $f = \frac{v}{\lambda} = \frac{3.0}{6} = 0.5 \text{ Hz}$</td></tr></table>	Amplitude	Frequency	Amplitude = $\frac{0.4}{2} = 0.2 \text{ m}$	Wavelength = $\frac{12 \text{ m}}{2} = 6 \text{ m}$ $f = \frac{v}{\lambda} = \frac{3.0}{6} = 0.5 \text{ Hz}$		
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22	E	69	☒ A wavelength should be same before and after barrier. ☒ B ends of waves should curve as waves diffract and be longer than gap in barrier. ☒ C boths sides of diffracted waves should be curved after the barrier ☒ D this answer is clearly nonsense ☒ E wavelength the same after barrier, ends of wave curve as wave diffracts in shadow area.						
23	A	41	☒ A Ray P has bent away from the normal as it has gone from a more dense medium to less dense ☒ B Ray Q has not changed direction and has continued in a straight line from the glass ☒ C Ray R is on wrong side of the normal ☒ D Ray S is on wrong side of the normal ☒ E Ray T is on wrong side of the normal						
24	E	65	$2.4 \times 10^4 \text{ Bq}$ $\therefore 1 \text{ s} = 2.4 \times 10^4 \text{ decays}$ $\therefore 15 \times 60 \text{ s} = 2.4 \times 10^4 \text{ decays} \times 15 \times 60 = 2.2 \times 10^7$						
25	B	62	<table><tr><th>Statement I - Incorrect</th><th>Statement II - Correct</th><th>Statement III - Incorrect</th></tr><tr><td>Nuclear <i>fission</i> is when a large nucleus splits into smaller nuclei</td><td>At the very hot temperatures required for fusion to take place, containment of plasma is an issue</td><td>Nuclear fusion takes place only at <i>high</i> temperatures</td></tr></table>	Statement I - Incorrect	Statement II - Correct	Statement III - Incorrect	Nuclear <i>fission</i> is when a large nucleus splits into smaller nuclei	At the very hot temperatures required for fusion to take place, containment of plasma is an issue	Nuclear fusion takes place only at <i>high</i> temperatures
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2b(iii)	1.4 m	Height dropped from = area under graph (1 mark) = $\frac{1}{2} \times 0.53 \times 5.2$ (1 mark) = 1.4 m (1 mark)																					
2c(i)	0.43 m s^{-2}	$F_{\text{un}} = 54 \text{ N} - 22 \text{ N} = 32 \text{ N}$ $m = 74 \text{ kg}$ $a = ?$ $F = m \times a$ $32 = 74 \times a$ $a = 0.43 \text{ m s}^{-2}$																					
2c(ii)	Reduces friction or air resistance	The cyclist take sthe full air resistamnce while the rider behind benefits from less air resistance as a result.																					
3a	Answer to include:	1 mark	Astra 1KR																				
		1 mark	it is a geostationary satellite or it has an orbital period of 24 hours or it is at an orbital altitude of 36 000 km																				
3b	27N	$W = ?$ $m = 3.5\text{kg}$ $g = 7.7 \text{ N kg}^{-1}$ $W = m \times g$ (1 mark) $W = 3.5 \times 7.7$ (1 mark) $W = 27 \text{ N}$ (1 mark)																					
3c	Answer is Greater then 101 minutes and Less than 676 minutes	<table><tr><td>Satellite</td><td>Orbital Altitude (km)</td><td>Orbital Period</td></tr><tr><td>UKube-1</td><td>825</td><td>101 minutes</td></tr><tr><td>Satellite</td><td>1200</td><td>Longer than 101minutes & shorter than 676 minutes</td></tr><tr><td>Kosmos2460</td><td>19100</td><td>676 minutes</td></tr></table>				Satellite	Orbital Altitude (km)	Orbital Period	UKube-1	825	101 minutes	Satellite	1200	Longer than 101minutes & shorter than 676 minutes	Kosmos2460	19100	676 minutes						
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4	Open Ended Question: <table><tr><th colspan="4">Candidate Marks</th></tr><tr><th>0 marks</th><th>1 mark</th><th>2 marks</th><th>3 marks</th></tr><tr><td>37%</td><td>39%</td><td>17%</td><td>6%</td></tr></table>	Candidate Marks				0 marks	1 mark	2 marks	3 marks	37%	39%	17%	6%	1 mark	2 marks	3 marks							
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		Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.	Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.	Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.																			
5a	One answer from:	not affected by weather	no (distortion from) atmosphere	no light pollution	can use telescope during the day																		
5b	Hydrogen Calcium (both required for 1 mark)	 All hydrogen lines in line spectra from star Some helium lines missing from star Some mercury lines missing from star All calcium lines in line spectra from star Some sodium lines missing from star																					
5c	working showing $3.2 \times 10^{18} \text{ m}$	$d = v \times t$ (1 mark) $d = 3.0 \times 10^8 \times 343 \times 365.25 \times 24 \times 60 \times 60$ (1 mark) $d = 3.2 \times 10^{18} \text{ m}$																					
6a	0.20 A	Total Resistance when Switch S_1 is closed = $36\Omega + 24\Omega = 60\Omega$ (1 mark) $V = 12 \text{ V}$ $I = ?$ $R = 60 \Omega$ $V = I \times R$ (1 mark) $12 = I \times 60$ (1 mark) $I = 0.20 \text{ A}$ (1 mark)																					
6b(i)	42Ω	Combine 2 parallel resistors		Combine with series resistor																			
		$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ (1 mark) $\frac{1}{R_T} = \frac{1}{36} + \frac{1}{36}$ (1 mark) $\frac{1}{R_T} = \frac{2}{36}$ $R_T = 18 \Omega$		<table><tr><td>R_T</td><td>=</td><td>R_1</td><td>+</td><td>R_2</td><td></td></tr><tr><td>R_T</td><td>=</td><td>18</td><td>+</td><td>24</td><td>(1 mark)</td></tr><tr><td>R_T</td><td>=</td><td>42Ω</td><td></td><td></td><td>(1 mark)</td></tr></table>		R_T	=	R_1	+	R_2		R_T	=	18	+	24	(1 mark)	R_T	=	42Ω			(1 mark)
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6b(ii)	Answer to include:	<table><tr><td>1 mark</td><td>(Ammeter reading will be) greater</td></tr><tr><td>1 mark</td><td>Total circuit resistance will be less</td></tr></table>		1 mark	(Ammeter reading will be) greater	1 mark	Total circuit resistance will be less		
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7a(i)	Graph showing:	<table><tr><td>1 mark</td><td>1 mark</td><td>1 mark</td></tr><tr><td>Suitable scales, labels and units</td><td>All points plotted accurately to $\pm$ half a division</td><td>Best fit <u>curve</u></td></tr></table>	1 mark	1 mark	1 mark	Suitable scales, labels and units	All points plotted accurately to $\pm$ half a division	Best fit <u>curve</u>	
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Suitable scales, labels and units	All points plotted accurately to $\pm$ half a division	Best fit <u>curve</u>							
7a(ii)	$74 \times 10^{-12} \text{ C}$								
7b	Any two from: (1 mark each)	Repeat measurements and average	Repeat measurements to identify outliers/rogue points	Increase the range of distances	Increase the number of different distances				
8a(i)	2120 J	$E_h = ?$ $E =$ $E =$ $E =$	$c = 532 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$ $= c$ $= 532$ $= 2120 \text{ J}$	$m = 1.90 \times 10^{-2} \text{ kg}$ $\times m$ $\times 1.90 \times 10^{-2}$	$\Delta T = 235^\circ\text{C} - 25^\circ\text{C} = 210^\circ\text{C}$ $\times \Delta T$ (1 mark) $\times 210$ (1 mark) (1 mark)				
8a(ii)	Heat (energy) lost to the surroundings.	Heat loss must be identified as being lost to the surroundings							
8b	Answer to include:	<table><tr><td>1 mark</td><td>1 mark</td></tr><tr><td>different (specific) heat capacity</td><td>different mass</td></tr></table>		1 mark	1 mark	different (specific) heat capacity	different mass		
1 mark	1 mark								
different (specific) heat capacity	different mass								
9a(i)	Force per unit area or Force per m^2	Pressure is calculated from equation $P = \frac{\text{Force}}{\text{Area}}$ and is defined as Force per unit area							

9a(ii)	$2.5 \times 10^{-3} \text{ m}^3$	$p_1 = 101 \text{ kPa}$ $p_2 = 92 \text{ kPa}$ $V_1 = 2.3 \times 10^{-3} \text{ m}^3$ $V_2 = ?$ (1 mark) $p_1 V_1 = p_2 V_2$ (1 mark) $101 \times 2.3 \times 10^{-3} = 92 \times V_2$ $\frac{101 \times 2.3 \times 10^{-3}}{92} = V_2$ (1 mark) $2.5 \times 10^{-3} \text{ m}^3 = V_2$																													
9a(iii)	The (gas) particles collide with the walls (of the crisp packet).	Pressure is caused by the particles inside a gas colliding with the walls of they container the gas in is. The faster te gas particles collide with the walls of the container the higher the pressure.																													
9b	$21 \text{ } \mu\text{Sv}$	$\dot{H} = 6.0 \text{ } \mu\text{Sv h}^{-1}$ $H = ?$ $t = 3.5 \text{ hours}$ $\dot{H} = \frac{H}{t}$ (1 mark) $6.0 = \frac{H}{3.5}$ (1 mark) $H = 21 \text{ } \mu\text{Sv}$ (1 mark)																													
10a	2.6 m	$d = ?$ $v = 240 \text{ m s}^{-1}$ $t = \frac{0.015\text{s}}{2} = 0.0075\text{s}$ (1 mark) $d = v \times t$ (1 mark) $d = 340 \times 0.0075$ (1 mark) $d = 2.6 \text{ m}$ (1 mark)																													
10b(i)	Working shown to calculate 45000Hz	$f = ?$ $N = 9 \text{ waves}$ $t = 2.0 \times 10^{-4} \text{ s}$ $f = \frac{N}{t} = \frac{9}{2.0 \times 10^{-4}} = 45000 \text{ Hz}$ (1 mark) (1 mark)																													
10b(ii)	One answer from:	<table><tr><td>Speed of sound in air is the same</td><td>The distance is the same</td></tr></table>	Speed of sound in air is the same	The distance is the same																											
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11a	P: Ultraviolet Q: Infrared	<table><tr><td>EM Type</td><td>Gamma</td><td>X-Ray</td><td>Ultra-violet</td><td>Visible</td><td>Infra-Red</td><td>Microwave</td><td>Radio & TV</td></tr><tr><td>Energy</td><td>High</td><td colspan="4"></td><td>Low</td></tr><tr><td>Frequency</td><td>High</td><td colspan="4"></td><td>Low</td></tr><tr><td>Wavelength</td><td>Low</td><td colspan="4"></td><td>High</td></tr></table>	EM Type	Gamma	X-Ray	Ultra-violet	Visible	Infra-Red	Microwave	Radio & TV	Energy	High					Low	Frequency	High					Low	Wavelength	Low					High
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11c(i)A	Working showing calculation of wavelength	$v = 3.0 \times 10^8 \text{ m s}^{-1}$ $f = 2.42 \text{ GHz} = 2.42 \times 10^9 \text{ Hz}$ $\lambda = ?$ $v = f \times \lambda$ (1 mark) $3.0 \times 10^8 = 2.42 \times 10^9 \times \lambda$ (1 mark) $\lambda = 0.12 \text{ m}$																													
11c(i)B	microwaves																														
11c(ii)A	$2.3 \times 10^{-5} \text{ J}$	$D = 5.0 \text{ } \mu\text{Gy} = 5.0 \times 10^{-6} \text{ Gy}$ $E = ?$ $m = 4.5 \text{ kg}$ $D = \frac{E}{m} \therefore 5.0 \times 10^{-6} = \frac{E}{4.5} \therefore E = 5.0 \times 10^{-6} \times 4.5 = 2.3 \times 10^{-5} \text{ J}$ (1 mark) (1 mark) (1 mark)																													

11c(ii)B	5.0x10 ⁻⁶ Sv	H = ? D = 5.0 μGy = 5.0x10⁻⁶ Gy H = D W_r (1 mark) H = 5.0x10⁻⁶ x 1 (1 mark) H = 5.0x10⁻⁶ Sv (1 mark) w_r (X-rays) = 1																					
12	Open Ended Question: <table><tr><th colspan="4">Candidate Marks</th></tr><tr><th>0 marks</th><th>1 mark</th><th>2 marks</th><th>3 marks</th></tr><tr><td>47%</td><td>32%</td><td>15%</td><td>6%</td></tr></table>	Candidate Marks				0 marks	1 mark	2 marks	3 marks	47%	32%	15%	6%	<table><tr><th>1 mark</th><th>2 marks</th><th>3 marks</th></tr><tr><td>Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.</td><td>Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.</td><td>Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.</td></tr></table>				1 mark	2 marks	3 marks	Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.	Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.	Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.
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13a	Answer to include:	<table><tr><td>1 mark</td><td colspan="4">Source Y</td></tr><tr><td>1 mark</td><td colspan="4">Sheet of paper absorbs alpha radiation (reducing the count rate)</td></tr><tr><td>1 mark</td><td>lead absorbs gamma radiation reducing the count rate further</td><td>Some gamma radiation is able to penetrate lead</td><td>The gamma radiation is able to penetrate the aluminium</td><td>(Source) Y is the only source with a reduction in count rate due to paper</td></tr></table>				1 mark	Source Y				1 mark	Sheet of paper absorbs alpha radiation (reducing the count rate)				1 mark	lead absorbs gamma radiation reducing the count rate further	Some gamma radiation is able to penetrate lead	The gamma radiation is able to penetrate the aluminium	(Source) Y is the only source with a reduction in count rate due to paper			
1 mark	Source Y																						
1 mark	Sheet of paper absorbs alpha radiation (reducing the count rate)																						
1 mark	lead absorbs gamma radiation reducing the count rate further	Some gamma radiation is able to penetrate lead	The gamma radiation is able to penetrate the aluminium	(Source) Y is the only source with a reduction in count rate due to paper																			
13b(i)	One answer from:	<table><tr><td>When an (uncharged) atom gains or loses electron(s)</td><td>When an (uncharged) atom gains electron(s)</td><td>When an (uncharged) atom loses electron(s)</td></tr></table>				When an (uncharged) atom gains or loses electron(s)	When an (uncharged) atom gains electron(s)	When an (uncharged) atom loses electron(s)															
When an (uncharged) atom gains or loses electron(s)	When an (uncharged) atom gains electron(s)	When an (uncharged) atom loses electron(s)																					
13b(ii)	Answer to include:	<table><tr><td>1 mark</td><td colspan="4">Alpha radiation</td></tr><tr><td>1 mark</td><td>Alpha radiation only has a short range (in air)</td><td>or</td><td colspan="2">Fewer alpha particles would reach the spark counter at increased distance</td></tr></table>				1 mark	Alpha radiation				1 mark	Alpha radiation only has a short range (in air)	or	Fewer alpha particles would reach the spark counter at increased distance									
1 mark	Alpha radiation																						
1 mark	Alpha radiation only has a short range (in air)	or	Fewer alpha particles would reach the spark counter at increased distance																				
13b(iii)	7.5x10 ⁻⁸ C	Q = ? I = 0.12 μA = 0.12x10⁻⁶ A t = 1 minute = 60s For 96 sparks: Q = I x t (1 mark) Q = 0.12x10⁻⁶ x 60 (1 mark) Q = 7.2x10⁻⁶ C For 1 spark: Q = $\frac{7.2 \times 10^{-6} \text{ C}}{96}$ (1 mark) Q = 7.5x10⁻⁸ C (1 mark)																					
14a(i)	background count rate	The background count rate should be subtracted from the count rate to the corrected count rate is a true measure of the activity from inside the boy.																					
14a(ii)	60 days	Count rate halves from 2000 to 1000 counts per minute Take any halving of the corrected count rate on the y-axis. Work out the time interval on the x-axis for this halving.																					

14a(iii)	180 days	$1 \xrightarrow{\quad} \frac{1}{2} \xrightarrow{\quad} \frac{1}{4} \xrightarrow{\quad} \frac{1}{8}$ 3 half-lives to decrease activity to $\frac{1}{8}$ of original value (1 mark) 1 half-life = 60 days $\therefore$ 3 half-lives = 3×60 days = 180 days (1 mark)			
14a(iv)	4000 counts per minute	Half-life is 60 days. $\therefore$ 0 days is one half-life before 60 days and the count rate at 0 days should be double the count rate at 60 days. At 60 days the count rate is 2000 counts per minute $\therefore$ at 0 days the count rate should be double at 4000 counts per minute.			
14b	Any suitable answer including:	tracers	sterilisation	smoke detectors	measuring thickness of paper

2023 National 5 Physics Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	E	Missing	Invalid
1	D	3	6	3	82	6	1	0
2	D	21	21	4	44	9	1	0
3	C	4	7	62	9	17	1	0
4	D	3	19	43	28	6	1	0
5	B	45	40	9	3	2	1	0
6	D	1	3	6	58	31	1	0
7	A	67	19	9	2	2	1	0
8	D	7	36	5	48	2	1	0
9	E	5	5	6	5	79	1	0
10	C	2	6	81	2	8	1	0
11	D	18	7	33	35	6	1	0
12	A	58	6	11	7	17	1	0
13	C	32	14	44	5	4	1	0
14	E	1	9	17	2	70	1	0
15	C	4	9	64	18	3	2	0
16	C	2	3	64	9	21	1	0
17	B	16	57	15	7	3	2	0
18	B	10	65	8	4	13	1	0
19	A	56	17	11	8	8	1	0
20	C	6	18	48	21	4	2	0
21	B	19	58	13	5	5	1	0
22	E	24	1	1	4	69	1	0
23	A	41	16	29	10	3	1	0
24	E	7	10	5	11	65	1	0
25	B	10	62	13	5	9	1	0

2023 Nat5 Physics Section 2 Marks Data Analysis

2023 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	2	1.3	65.7%	58	4.3%
Q1aii	2	0.8	42.5%	64	9.5%
Q1bi	3	1.5	50.2%	56	4.1%
Q1bii	2	0.6	30.1%	59	22.3%
Q1ci	2	1.8	88.1%	46	3.2%
Q1cii	3	1.5	50.0%	62	12.6%
Q1ciii	1	0.1	11.7%	34	11.4%
Q2a	1	0.3	31.1%	36	5.0%
Q2bi	3	1.2	39.3%	63	10.5%
Q2bii	2	0.7	36.8%	55	11.9%
Q2biii	3	0.8	25.1%	51	21.8%
Q2ci	4	3	74.9%	61	5.2%
Q2cii	1	0.6	61.5%	36	7.0%
Q3a	2	1.2	59.8%	31	2.4%
Q3b	3	2.6	86.3%	52	2.4%
Q3c	1	0.8	75.5%	29	6.0%
Q4	3	0.9	30.8%	53	6.0%
Q5a	1	0.5	49.7%	37	4.2%
Q5b	1	0.8	78.4%	30	1.0%
Q5c	2	0.7	36.7%	44	9.0%
Q6a	4	2.7	68.1%	61	7.6%
Q6bi	4	2.4	60.9%	62	7.4%
Q6bii	2	0.8	41.1%	51	8.3%
Q7ai	3	1.8	59.5%	47	3.7%
Q7aii	1	0.5	51.6%	44	7.3%
Q7b	2	0.6	31.1%	43	13.5%
Q8ai	3	2.1	70.9%	60	6.1%
Q8aii	1	0.5	45.9%	48	9.5%
Q8b	2	1.2	58.5%	57	6.3%
Q9ai	1	0.2	18.7%	41	8.3%
Q9aii	3	2	68.2%	62	9.6%
Q9aiii	1	0.3	26.0%	44	28.3%
Q9b	3	1.4	48.3%	66	9.1%
Q10a	4	2	49.7%	65	7.6%
Q10bi	2	1.3	66.3%	50	10.8%
Q10bii	1	0.5	51.1%	51	19.1%
Q11a	1	0.7	72.9%	41	2.9%
Q11b	2	1.2	60.4%	49	5.7%
Q11ciA	2	1.4	67.7%	69	10.7%
Q11ciB	1	0.6	57.8%	54	6.8%
Q11ciiA	3	2.2	72.1%	66	8.3%
Q11ciiB	3	1.7	57.1%	68	15.8%
Q12	3	0.8	26.5%	63	18.0%
Q13a	3	1.3	44.1%	53	7.0%
Q13bi	1	0.4	40.7%	44	12.0%
Q13bii	2	0.8	39.7%	53	13.8%
Q13biii	4	1.7	43.6%	65	21.8%
Q14ai	1	0.2	16.5%	42	19.7%
Q14aii	1	0.3	32.3%	47	10.9%
Q14aiii	2	0.9	43.7%	60	17.8%
Q14aiv	1	0.2	20.5%	39	20.7%
Q14b	1	0.3	30.7%	40	11.4%
Paper	110	56.8	51.7%		

2023 Nat5 Physics Section 2 Distribution of Marks Analysis

2023 Item	0	1	2	3	4
Q1ai	30	8	62		
Q1aii	42	32	26		
Q1bi	32	3	48	17	
Q1bii	61	18	21		
Q1ci	7	10	83		
Q1cii	44	5	7	43	
Q1ciii	88	12			
Q2a	69	31			
Q2bi	48	18	2	32	
Q2bii	50	27	23		
Q2biii	73	3	2	23	
Q2ci	11	14	2	12	61
Q2cii	39	61			
Q3a	37	6	57		
Q3b	4	4	19	72	
Q3c	24	76			
Q4	37	39	17	6	
Q5a	50	50			
Q5b	22	78			
Q5c	53	20	27		
Q6a	16	19	2	4	60
Q6bi	33	2	4	10	51
Q6bii	46	26	28		
Q7ai	14	15	48	22	
Q7aii	48	52			
Q7b	50	37	12		
Q8ai	10	12	32	46	
Q8aii	54	46			
Q8b	22	38	39		
Q9ai	81	19			
Q9aii	19	6	24	50	
Q9aiii	74	26			
Q9b	29	21	27	23	
Q10a	26	17	22	2	33
Q10bi	27	13	60		
Q10bii	49	51			
Q11a	27	73			
Q11b	33	13	54		
Q11ciA	18	29	53		
Q11ciB	42	58			
Q11ciiA	13	7	30	50	
Q11ciiB	23	22	14	40	
Q12	47	32	15	6	
Q13a	39	19	14	29	
Q13bi	59	41			
Q13bii	50	20	30		
Q13biii	31	12	30	6	21
Q14ai	83	17			
Q14aii	68	32			
Q14aiii	51	10	39		
Q14aiv	80	20			
Q14b	69	31			